Student ID (SID):
Student's Name:

CSCE-4613/5613 Artificial Intelligence

Midterm Exam

Sample Questions

Time: 12:30 p.m – 1:45 p.m

Instructions

1. It is an individual exam.
2. You have 75 minutes. You may not leave during the last 10 minutes of the exam.
3. Do NOT open exams until told to. Write your SIDs in the top left corner of every page.
4. If you need to go to the bathroom, bring us your exam, phone, and SID. We will record the time.
5. In the interest of fairness, we want everyone to have access to the same information. To that end, we will not be answering questions about the content.
6. The exam is closed book, closed laptop, and closed notes except your ONE-page double-sided cheat sheet. You are allowed a non-programmable calculator for this exam. Turn off and put away all other electronics.
7. Mark your answers - ON THE EXAM IN THE ANSWER AREAS. We will not grade anything on scratch paper.

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Question 1: What is the rank of the following matrix? (1 points)

$$\begin{bmatrix} 1 & -2 & 0 & 4 \\ 3 & 1 & 1 & 0 \\ -1 & -5 & -1 & 8 \\ 3 & 8 & 2 & -12 \end{bmatrix}$$

Choose ONE of these answers

(a) A: 0 (b) B: 1 (c) C: 2 (d) D: 3 (e) E: 4

Question 2: What is the rank of the following matrix? (1 points)

$$\begin{bmatrix} 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \end{bmatrix}$$

Choose ONE of these answers

(a) A: 0 (b) B: 1 (c) C: 2 (d) D: 3 (e) E: 4

1: 30%, 3% loss
2: 20%, 5% loss
3: 50%, 1% loss
.3*.03 + .2*.05 + .5*.01 = .024 = 2.4%

Question 3: (2 points)

The entire output of a factory is produced on three machines. The three machines account for 30%, 20%, and 50% of the factory output. The fraction of defective items produced is 3% for the first machine; 5% for the second machine; and 1% for the third machine.

What is the probability that a randomly chosen item produced in this factory is defective?

Choose ONE of these answers

(a) 2.4%

(b) 10%

(c) 12.5%

(d) 20.8%

(e) None of above

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Question 4: (2 points)

The entire output of a factory is produced on three machines. The three machines account for 30%, 20%, and 50% of the factory output. The fraction of defective items produced is 3% for the first machine; 5% for the second machine; and 1% for the third machine.

If an item is chosen at random from the total output and is found to be defective, what is the probability that it was produced by the third machine?

Choose ONE of these answers

(a) 50%

(b) 15%

(c) 12.5%

(d) 20.8%

(e) None of above

1: 30%, 3% loss
2: 20%, 5% loss
3: 50%, 1% loss
(.5*.01)/(.024) = 0.2083

Question 5: (1 points)

Choose ONE of these answers

In a Search algorithm, the measures of performance include:

(a) Completeness

(b) Optimality

(c) Time

(d) Space

(e) All of above

Question 6: (1 points)

Choose ONE of these answers

In a Search algorithm, *Completeness* means

(a) Guarantee to find a solution when there is one

(b) Find an optimal solution

(c) How long does it take to find a solution

(d) How much memory is needed to conduct the search

(e) None of above

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Question 7: (1 points)

Choose ONE of these answers

In a Search algorithm, *Optimality* means

(a) Guarantee to find a solution when there is one.

(b) Find an optimal solution

(c) How long does it take to find a solution

(d) How much memory is needed to conduct the search

(e) None of above

Question 8: (1 points)

Choose ONE of these answers

In a Search algorithm, the search cost is about

(a) Time cost

(b) Space cost

(c) Path cost

(d) Both (a) and (b)

(e) All of above

Question 10: (1 points)

Choose ONE of these answers
Uninformed Search algorithms include:

(a) A Search
(b) A* Search
(c) Greedy (Best-First) Search
(d) Both (a) and (b)
(e) All of above

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Question 11: (1 points)

Choose ONE of these answers

In Breadth-first Search:

(a) Guarantee to find a solution if one exists

(b) Guarantee to find the least cost path

(c) Don't guarantee to find a solution if one exists

(d) Don't guarantee to find the least cost path

(e) Both (a) and (b)

Question 12: (1 points)

Choose ONE of these answers

In Depth-first Search:

(a) Guarantee to find a solution if one exists

(b) Don't guarantee to find a solution if one exists

ALSO D) SO IDK WHATS UP

(c) Guarantee to find the least cost path

(d) Don't guarantee to find the least cost path

(e) Both (a) and (c)

(f) Both (a) and (d)

Question 13: (1 points)

When will Breadth-first Search (BFS) outperform Depth-first Search (DFS)?

Answer: When the goal is not deep (goal is shallow/close to start), or when you want the shortest path in number of steps

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Question 14: (1 points)

When will Depth-first Search (DFS) outperform Breadth-first Search (BFS)?

Answer:

When answer is deep and far from start, and when memory is limited.

Question 15: (1 points)

Consider the search problem represented in Figure 1, where *P* is the start node. We want to run DFS to visit all nodes in an order. Which sequences of paths are explored by DFS?

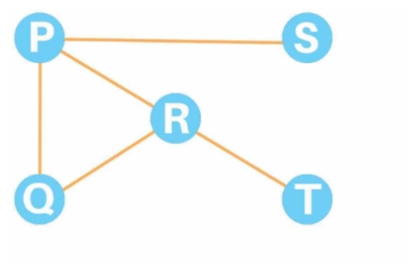

Figure 1. Input Graph

Choose ONE of these answers

(a) P Q R S T
(b) P Q R T S
(c) P R S Q T
(d) None of them

Question 16: (1 points)

Consider the search problem represented in Figure 1, where *P* is the start node. We want to run BFS to visit all nodes in order. Which sequences of paths are explored by BFS?

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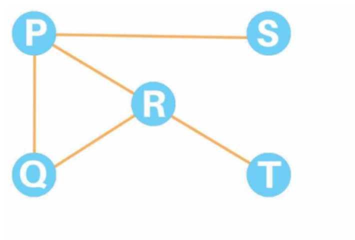

Figure 2. Input Graph

Choose ONE of these answers

(a) P Q R S T
(b) P Q R T S
(c) P R S Q T
(d) None of them

Question 17: (1 points)

In Search Heuristic, choose ONE of these answers:

(a) A heuristic is a function that estimates how far a state is from the starting point.
(b) A heuristic is a function that estimates how close a state is to a goal
(c) Both (a) and (b)
(d) None of above

Question 18: (1 points)

In Search Heuristic, choose ONE of these answers:

(a) A heuristic is designed for a particular search problem
(b) A heuristic can be applied for both Uniformed and Informed search algorithms.
(c) A heuristic is designed for Informed search algorithms
(d) Both (a) and (c)
(e) None of above

Question 19: (1 points)

In Uniform-Cost Search (UCS), choose ONE of these answers:

(a) UCS is an extension of BFS to the weighted graph case.

(b) UCS is complete

(c) UCS is optimal

(d) All of above

(e) None of above

Question 20: (1 points)

When UCS is equivalent to BFS, choose ONE of these answers:

(a) If all edges have different cost

(b) If the path has a ring

(c) Both (a) and (b)

(d) None of them

UCS = BFS IF ALL EDGES HAVE THE SAME COST

Question 21: (2 points)

In Uniform-Cost Search in Figure 2, write the searching path and the cost of the solutions (start: s, end: g) (final step):

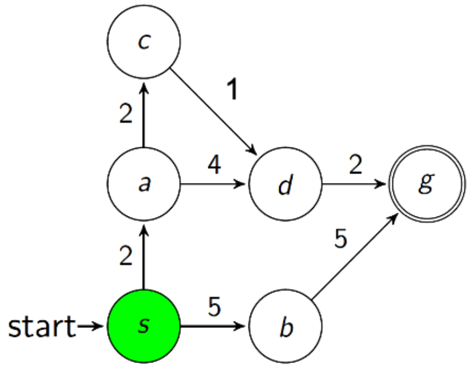

Figure 2

Path	Cost
s	0
sa	2
sac	4
sacd	5
sacdg	7

Question 22: (2 points)

In Greedy (Best-First) Search in Figure 3, write the searching path, the cost and h of the solutions (start: s, end: g) (final step):

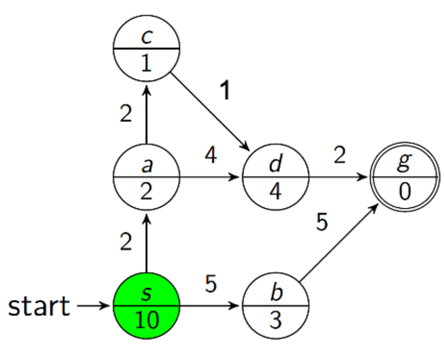

Figure 3

Path	Cost	h

s	0	10
sb	5	3
sbg	10	0

Question 23: (1 points)

In Greedy (Best-First) Search, choose ONE of these answers:

(a) Greedy search is complete

(b) Greedy search is optimal

(c) Greedy search is complete but not optimal

(d) None of above

neither complete nor optimal

Question 24: (1 points)

In Greedy (Best-First) Search, choose ONE of these answers:

(a) It is often fast and efficient

(b) It depends on the heuristic function

(c) It belongs to Uninformed Search

(d) All of above

(e) Both (a) and (b)

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Question 25: (1 points)

The advantage of Greedy (Best-First) Search compared to Uniform-Cost Search (UCS), choose ONE of these answers:

(a) Explore paths with all directions

(b) Minimize the estimated distance to the goal

(c) It has the heuristic function

(d) All of above

(e) Both (b) and (c)

Question 26: (1 points)

In Greedy (Best-First) Search, choose ONE of these answers:

(a) Worst-case time and space complexity: O(bm)

(b) Worst-case time and space complexity: O(b^m)

(c) Worst-case time and space complexity: infinite

(d) None of above

Question 27: (1 points)

In Greedy (Best-First) Search, choose ONE of these answers:

(a) It is similar in spirit to BFS

(b) It is similar in spirit to DFS

(c) It is similar in spirit to BFS with a heuristic function

(d) None of above

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Question 28: (2 points)

In A Search in Figure 4, write down the searching path and the cost, h and f of the solutions (start: s, end: g) (final step):

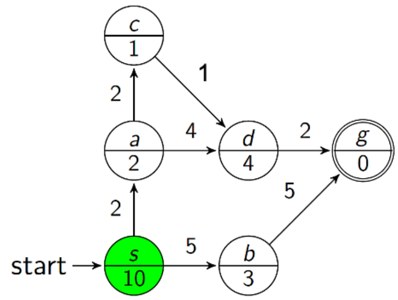

Figure 4

Path	Cost g	h	f
s	o	10	10
sa	2	2	4
sac	4	1	5
sacd	5	4	9
sacdg	7	0	7

Question 29: (1 points)

In A Search, choose ONE of these answers:

(a) A Search is similar to UCS

(b) A Search is similar to Greedy (Best-First) Search

(c) A Search is complete

(d) A Search is optimal

(e) Both (c) and (d)

Question 30: (1 points)

In A Search, choose ONE of these answers:

(a) A Search is not equivalent to UCS when the heuristic h = 0

(b) A Search is equivalent to UCS when the heuristic h = 0

(c) A Search is equivalent to UCS when the heuristic $h \neq 0$

(d) None of above

Question 31: (1 points)

In A Search, choose ONE of these answers:

(a) A Search is optimal

(b) A Search includes an admissible heuristic

(c) A Search is not optimal

(d) Answers are (a) and (b)

Question 32: (1 points)

In A* Search, choose ONE of these answers:

(a) A* Search is not complete

(b) A* Search is optimal

(c) A* Search is complete but not optimal

(d) Answers are (a) and (b)

Question 33: (2 points)

In A* Search in Figure 5, write down the searching path and the cost, h and f of the solutions (start: s, end: g) (final step):

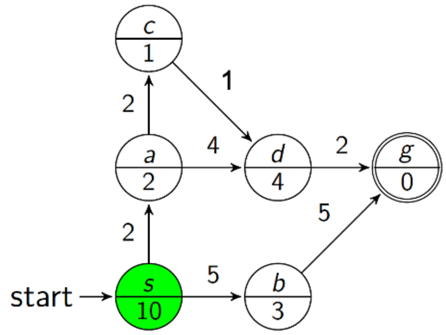

Figure 5

Path	Cost g	h	f
s	0	10	10
sa	2	2	4
sac	4	1	5
sacd	5	4	9
sacdg	7	0	7

Question 34: (1 points)

In Constraint satisfaction problems (CSPs), choose ONE of these answers:

(a) The same as standard search algorithms

(b) Allow useful general-purpose algorithms

(c) Both (a) and (b)

(d) None of above

Question 36: (1 points)

In Constraint satisfaction problems (CSPs), choose ONE of these answers:

(a) There is backtracking search in CSPs

(b) Depth-first search for CSPs with single-variable assignments is called backtracking search

(c) Breadth-first search for CSPs with single-variable assignments is called backtracking search

(d) Both (a) and (b)

(e) None of above

Question 37: (1 points)

Choose ONE of these answers:

(a) N-Queens problem can be solved by Greedy (Best-First) Search

(b) N-Queens problem can be solved by CSPs

(c) Both (a) and (b)

(d) None of above

Question 38: (1 points)

Standard Search v.s. CSPs, Choose the best ONE of these answers:

(a) In Standard Search problem, state is a black box.

(b) In CSPs, state defined by variables, with values in a domain.

(c) In CSPs, goal test is a set of constraints.

(d) All of above

Question 39: (1 points)

On a six-sided dice, each side has a number between 1 and 6.

What is the probability of throwing a 3 or a 4?

Choose the best ONE of these answers:

(a) 1 in 6

(b) 1 in 3

(c) 1 in 2

(d) 1 in 4

(e) None of above

Question 40: (1 points)

Three coins are tossed up in the air, one at a time.

What is the probability that two of them will land heads up and one will land tails up?

Choose the best ONE of these answers:

(a) 0

(b) 1/8

(c) 1/4

(d) 3/8

(e) None of above

total outcomes = 2^3=8
good outs: HHT, HTH,
THH
so 3 out of 8 outcomes
are favorable

Question 41: (1 points)

A bag contains 14 blue, 6 red, 12 green, and 8 purple buttons. 25 buttons are removed from the bag randomly.

How many of the removed buttons were red if the chance of drawing a red button from the bag is now 1/3?

Choose the best ONE of these answers:

(a) 0

(b) 1

(c) 3

(d) 5

(e) 6

14B,6R, 12R, 8P
total = 40
P(R) = 1/3 after
removing 25
how many remove?
currently chance is
6/40
new total = 40 - 25=15
5/15 = 1/3 chance
which is 6 - 1 so
removing one red
marble would cause it
to be this 1/3 is red

Question 42: (1 points)

In a lottery game, there are 2 winners for every 100 tickets sold on average.

If a man buys 10 tickets, what is the probability that he is a winner?

Choose the best ONE of these answers:

(a) 21.5%

(b) 20%

(c) 18.3%

(d) 2%

1/50 chance of win ==
.02 probability
so probability of losing
is (1 - .02)^10 ==
.98^10 == .817
so one win at least:
1 - .817 = .183

(e) None of above

Question 43: (4 points)

The random variable X has a range of $\{0, 1, 2\}$ and the random variable Y has a range of $\{1, 2\}$. The joint distribution of X and Y is given in the following table:

x	y	$P(X = x, Y = y)$
0	1	0.2
0	2	0.1
1	1	0.0
1	2	0.2
2	1	0.3
2	2	0.2

1. Write down tables for the marginal distributions of X and of Y, i.e. give the values of P(X = x) for all x, and of P(Y = y) for all y. (1 points)

P(X):
x=0 => .2 + .1 = .3
x=1 => 0.0 + 0.2 = .2
x=2 => .3 + .2 = .5

P(Y):
Y=1 → 0.2 + 0.0 + 0.3
= 0.5
Y=2 → 0.1 + 0.2 + 0.2
= 0.5

2. Write down a table for the conditional distribution of X given that Y = 2, i.e. give the values of P(X = x | Y = 2) for all x. (1 points)

P(X = 0 | Y = 2) = .1 / .5 = .2
P(X = 1 | Y = 2) = .2 / .5 = .4
P(X = 2 | Y = 2) = .2 / .5 = .4

3. Compute E(X) and E(Y) (1 points)

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E(X)= 0(.3) + 1(.2) + 2(.5) = 1.2
E(Y)= 1(.5) + 2(.5) = 1.5

4. Compute E(XY) (1 points)

0 + 0 + 0 + .4 + .6 + .8 = 1.8

Question 44: (3 points)

You roll one red die and one green die. Define the random variables X and Y as follows:

X = The number showing on the red dice

Y = The number of dice that show the number six

For example, if the red and green dice show the numbers 6 and 4, then X = 6 and Y = 1.

1. Write down a table showing the joint probability mass function for X and Y (1 point)

x, y=0, y=1, y=2
1-5, 5/36, 1/36, 0
6, 0, 5/36, 1/36

2. Find the marginal distribution for Y (1 point)

P(Y = 0) = 25/36, P(Y = 1) = 10/36, P(Y = 2) = 1/36

3. Compute E(Y). (1 points)

E(Y) = 10/36 + 2/36 +
12/36 = 1/3